

Working with Visualizations

30-Minute Hands-On Exercise | Week 9 - Section 5A

Learning Objectives: Identify the key visualization types in Power BI and their best use cases | Configure visual field wells correctly for a given chart type | Apply accessibility and audience considerations when choosing a visual | Select the most appropriate chart type for a real analyst scenario | Explain what non-data visual elements contribute to a report

Instructions: Complete all four parts in 30 minutes. Refer to your Section 5A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 5A.

1. The visual field well that adds extra information visible when a user hovers over a data point is called a _____.
2. The field well that creates multiple parallel versions of the same chart -- one per category value -- is called _____.
3. A chart that displays relative values as rectangles inside a larger rectangle, arranged from largest (top left) to smallest (bottom right), is called a _____.
4. A chart that plots data points against two numerical axes and is useful for showing clusters or correlations in data is called a _____.
5. The visualization type that displays progress made toward a key performance indicator defined by the business is called a _____.
6. Images, text boxes, navigation buttons, and other non-data elements can be added to a report from the _____ in the Report view.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each visualization type.

Visual Type	Description / Best Use
A. Bar / Column Chart	1. Displays a single key data point or a small set of specific values prominently
B. Line / Area Chart	2. Traces values over time to show trends; the area version fills in the shape to emphasize magnitude of change

C. Pie / Donut Chart	3. Shows breakdown and subtotals of data across multiple row and column dimensions
D. Bubble Chart	4. Extends a scatter chart by sizing each data point based on an additional dimension of value
E. Matrix	5. Simple and powerful for comparing values across categories; sub-values can be stacked or clustered
F. Card	6. Compares parts within a whole; all slices must add up to 100% with no overlap

Part 3: Visual Field Configuration (~8 minutes)

For each chart type, identify what kind of data belongs in each field well. Use: Categorical, Numeric, Optional, or N/A. The first row is completed as an example.

Chart Type	X-Axis	Y-Axis / Values	Legend	Data Type (Cat. or Num.)
Bar / Column Chart	Categorical	Numeric	Optional	<i>X=Cat, Y=Num</i>
Line / Area Chart				<i>fill in above</i>
Pie / Donut Chart				<i>fill in above</i>
Scatter Chart				<i>fill in above</i>
Matrix				<i>fill in above</i>
Card				<i>fill in above</i>

Hint: Review Section 5A.2. X-axis is often categorical/qualitative; Y-axis or Values is often numeric/quantitative. Legend is typically categorical. For charts without a traditional X-axis (like Card or Pie), use N/A or describe the equivalent field.

Part 4: Scenario Analysis (~9 minutes)

Read each scenario. Select the most appropriate Power BI visualization type from Section 5A and briefly explain your reasoning in the right column.

Scenario	Best Visual Type + Why
1. A sales manager wants to see how monthly revenue has changed over the past 12 months to identify a trend.	

2. A marketing analyst needs to compare total ad spend across six different campaign channels for Q3.	
3. A finance team wants to show each department's share of the total annual budget as a proportion of the whole.	
4. An executive dashboard needs to display the single most important metric front and center: total year-to-date revenue.	
5. A data scientist wants to explore whether there is a relationship between customer age and total purchase amount, and whether clusters exist.	
6. A store operations manager wants to track whether weekly sales are on pace to hit the monthly target of \$500,000.	
7. A logistics analyst needs to visualize the number of deliveries made in each city across the country.	
8. A product manager needs to show sales broken down by both product category and sales region, including row and column subtotals.	

Bonus / Reflection (if time allows)

Section 5A states: 'It is very common to discover issues with your data once you start creating visualizations -- just consider that part of the process.' In 3-5 sentences, describe a specific example of a data issue that a visualization might reveal. What would the visual look like, what would it tell you is wrong, and what step of the modeling process would you return to in order to fix it?